FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – XI
PAPER –2
TEST DATE: 19-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. $\int Ndt = mv_y - mu_y, e = \frac{v_y}{u_y} \Rightarrow v_y = 0.4 \times 10 = 4$ m/s

$\int Ndt = m \times 4 - (-10m) = 14\ m$

$-\int \mu Ndt = mv_x - mu_x$

$-0.5 \times 14m = mv_x - m \times 10 \Rightarrow v_x = 3m/s$

2. A

Sol. $b = H\frac{\mu_0 iR^2}{2(R^2 + x^2)^{3/2}} = n\frac{\mu_0 i \sin^{3/2}\theta\sqrt{R}}{2}$

$\Rightarrow B \propto \frac{1}{R}$

3. A

Sol. $E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$

4. A

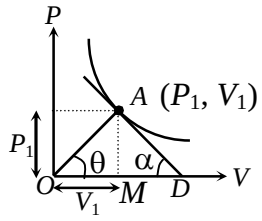
Sol. Decrease in K.E. = Increase in magnetic energy in inductor $\frac{1}{2}mv^2 - \frac{1}{2}m\left(\frac{v}{2}\right)^2 = \frac{1}{2}CIL$

$\frac{3}{4}\frac{mv^2}{L} = I^2$

Where mv = J

5. B, C

Sol. $\tan\alpha = -$ [slope at (P_1, V_1)] $= \dfrac{P_1}{V_1}$



$\tan\theta = \dfrac{P_1}{V_1}$

$\therefore \quad \alpha = \theta$

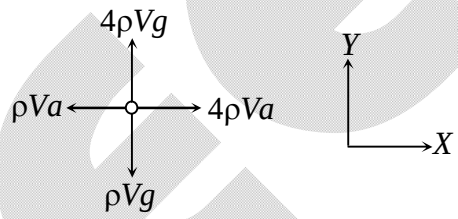
$OM = MD = V_1$

$\therefore \quad OD = 2V_1$

Area of triangle $AOD = \dfrac{1}{2}(AM \times OD) = \dfrac{1}{2}(P_1 \times 2V_1) = P_1V_1 = nRT$

6. B, C

Sol. F.B.D. of ball in frame of container



$a_y = \dfrac{4\rho Vg - \rho Vg}{\rho V} = 3g$

$\dfrac{L}{2} = \dfrac{1}{2}3gt^2 \Rightarrow t = \sqrt{\dfrac{L}{3g}}$

$a_x = \dfrac{4\rho Va - \rho Va}{\rho V} = 3a$

$L = \dfrac{1}{2}3at^2 \Rightarrow t = \sqrt{\dfrac{2L}{3a}}$

Ball to collide at point Q, $t = \sqrt{\dfrac{2L}{3a}} = \sqrt{\dfrac{L}{3g}} \Rightarrow a = 2g$

7. A, C

Sol. Velocity of A just after collision is v_0 in –ve x-axis and velocity of B is $a\omega$.

and $v_0 = A\omega = A\sqrt{\dfrac{k}{m}}$, $A = v_0\sqrt{\dfrac{m}{k}}$

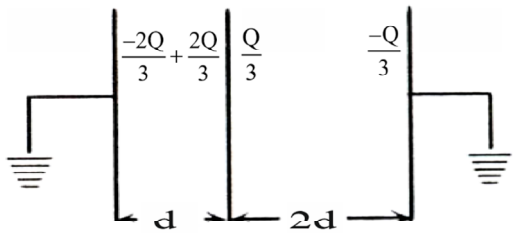
So equation is $x = x_0 - v_0\sqrt{\dfrac{m}{k}}\sin\omega t$

SECTION – B

8. 7

Sol. Energy stored $= \dfrac{q^2}{2C}$

$= \dfrac{4Q^2d}{9(2)\varepsilon_0 A} = \dfrac{2}{9}\dfrac{Q^2d}{\varepsilon_0 A}$



9. 6

Sol. $\dfrac{9\varepsilon^2(t)}{R} = mS\Delta T$ (d = density)

$$\frac{9\varepsilon^2 t}{\rho\ell/A} = (d)A\ell S\Delta T$$

$$\frac{N^2\varepsilon^2}{\frac{\rho(2\ell)}{A}}t = (d)A(2\ell)S\Delta T$$

$$36\varepsilon^2 = N^2\varepsilon^2$$

$$N = 6$$

10. 8

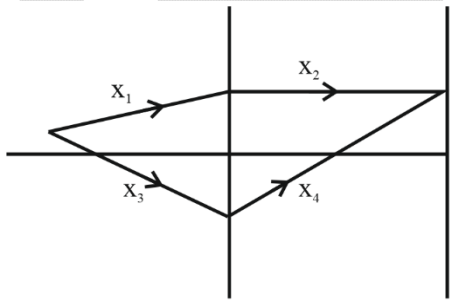
Sol. $r = \frac{mv}{qB}$, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $m = \frac{v}{u}$

11. 0

Sol. $x_4 - x_2 = \sqrt{D^2 + d^2} - D$

$$= \frac{d^2}{2D} = \frac{\lambda}{2}$$

Hence for t = 0, S is at central position and path difference is $\frac{\lambda}{2}$



12. 6

Sol. $\frac{h}{\sqrt{2km}} = \frac{hc}{eV}$

$$K = \frac{e^2v^2}{2mc^2}$$

where m = mass of electron

13. 5

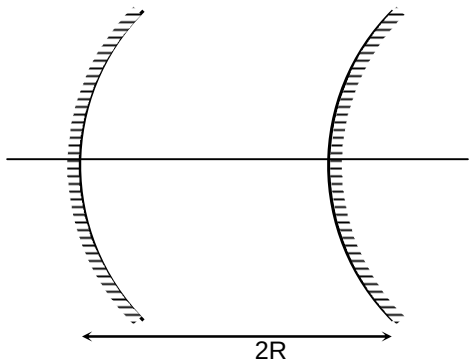
Sol. $\frac{1}{V_1} + \frac{1}{-2R} = \frac{-1}{R/2}$

$$\frac{1}{V_1} = \frac{-2}{R} + \frac{1}{2R} \Rightarrow V_1 = \frac{-2R}{3}$$

$$U_2 = 2R - \frac{2R}{3} = \frac{4R}{3}$$

$$\frac{1}{V_2} + \frac{3}{-4R} = +\frac{2}{R} \Rightarrow V_2 = \frac{4R}{11}$$

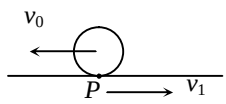
Magnification $m = m_1m_2$



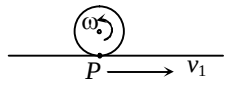
SECTION – C

14. 0.80

Sol. Angular momentum is conserved about a point *P*



Before



After

$$mv_0R = \frac{2mr^2\omega}{5}$$

$$v_0 = \frac{2}{5}v_1$$

Retardation = μg

$$v_0 = \mu g t_1$$

$$t_1 = \frac{2}{5}\frac{v_1}{\mu g} = 0.8 \text{ sec}$$

15. 3.84

Sol. Displacement of sphere x_1 towards left

$$v_0^2 = 2\mu g x_1$$

$$x_1 = \frac{2}{25}\frac{v_1^2}{\mu g}$$

Displacement of belt in time t_1, $x_2 = v_1 t_1 = \frac{2}{5}\frac{v_1^2}{\mu g}$

Relative displacement = $x_1 + x_2 = \frac{12}{25}\frac{v_1^2}{\mu g} = 3.84$ M

16. 0.20

Sol. Intensity of reflected wave is 64% of incident wave

$$\frac{A_R}{A_I} = \sqrt{\frac{I_R}{I_I}} = 0.8$$

Amplitude of reflected wave $A_R = 0.8A$

Given resultant wave is constituted by the following waves.

$$y = 0.8A\cos(bt + ax) - 0.8A\cos(bt - ax) + 0.2A\cos(bt + ax)$$

First two waves constitute stationary wave

Hence $c = 0.2$

17. 1.80

Sol. $v_{max} = \omega A$

Amplitude at position of antinode = A + 0.8 A

Hence $v_{max} = b(1.8A)$ (here $\omega = b$) .

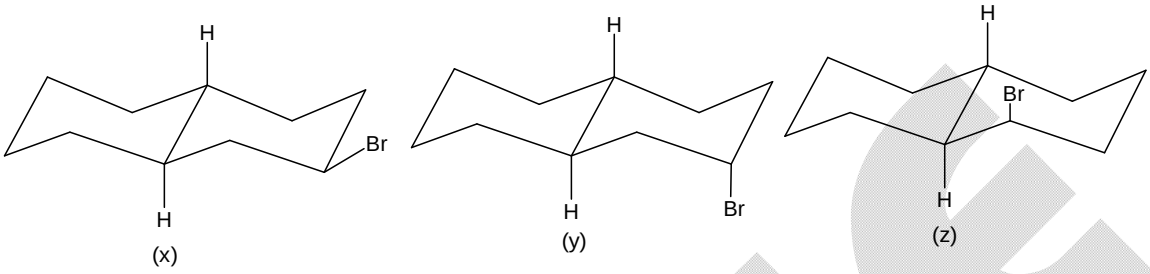
Chemistry

PART – II

SECTION – A

18. B

Sol.



Molecule (x) show slowest E2 reactivity as Br is present at equatorial position.

19. C

Sol. Conceptual

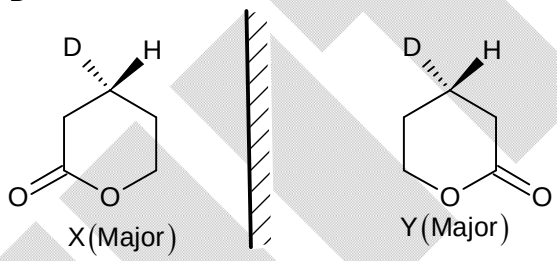
20. C

Sol.

	B. P. (^oC)	M. P. (^oC)
H_2O	100	0
NH_3	– 33.3	– 77.7
H_2SO_4	337	10
HF	19.5	– 83.6

21. D

Sol.



22. D

Sol. $i_{FeCl_3} = 1+(1+3-1)\times 0.8 = 3.4$

$\pi_A = 3.4\times 0.01\times R\times 300 = 10.2R$

$i_{K_4Fe(CN)_6} = 1+(4+1-1)\times 0.6 = 3.4$

$\pi_B = 3.4\times 0.1\times R\times 300 = 102R$

$\frac{\pi_B}{\pi_A} = 10;$ $\pi_B > \pi_A$ (B is hypertonic)

Blue colouration doesn't appear in any side as only solvent can flow through SPM.

23. B, D

Sol. $\Delta S_{isothermal} = 1\times R\times \ell n\left(\frac{2V}{V}\right) = R\ell n2$

$\Delta S_{isochoric} = 1 \times C_V \times \ell n\left(\frac{600}{300}\right) = C_V \ell n2$

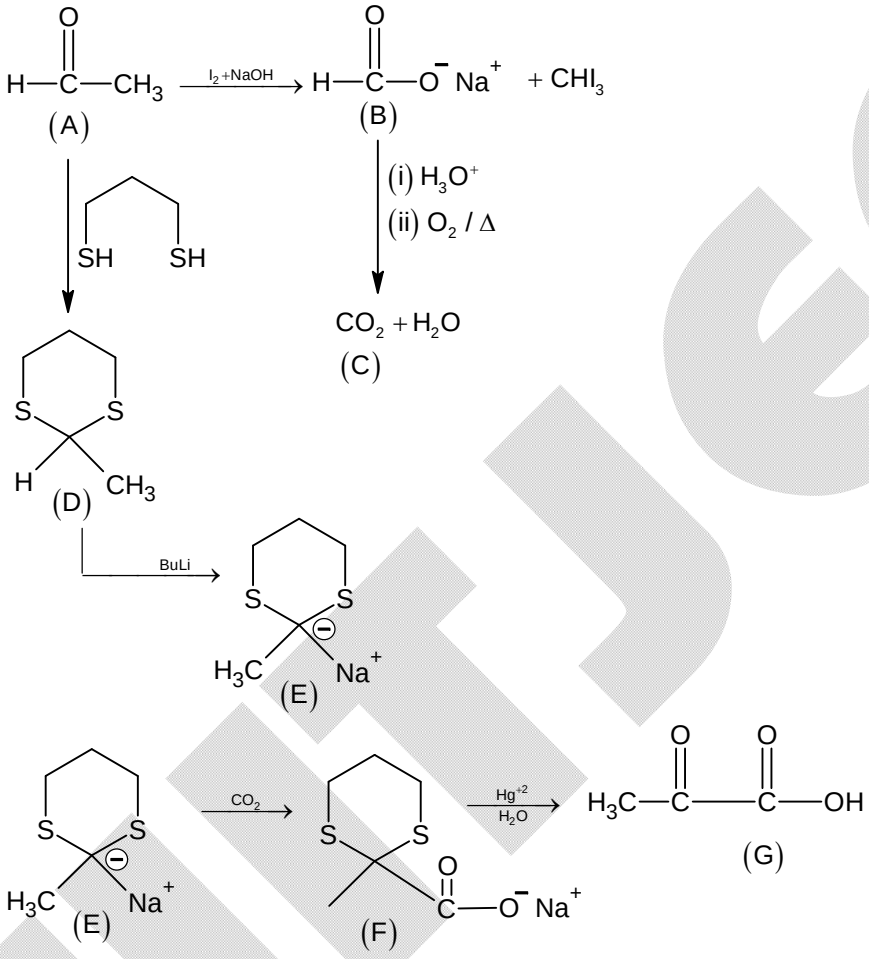
$\Delta S_{total} = R\ell n2 + C_V \ell n2$

$= (R + C_V)\ell n2$

$C_P \ell n2$

24. A, B, C

Sol.



SECTION – B

25. 18

Sol.

A	+	CH_3MgBr	$\longrightarrow$	$5CH_4$ + Salt
0.84 mol		(excess)		0
0				4.2 mol

On complete combustion of 4.2 mole of methane heat produced $= 4.2 \times 900 \times 10^3$ J

$q = m.s.\Delta T$

$4.2 \times 900 \times 10^3 = 50 \times 10^3 \times 4.2 \times \Delta T$

ΔT = 18

26. 100

Sol. At 300 K,

$\pi = iCRT$

$\Rightarrow i = \dfrac{\pi}{CRT} = \dfrac{3}{0.1 \times \dfrac{1}{12} \times 300} = 1.2$

$\therefore \alpha = 0.2$

At 360 K,

$i = \dfrac{\pi}{CRT} = \dfrac{4.2}{0.1 \times \dfrac{1}{12} \times 360} = 1.4$

$\therefore \alpha = 0.4$

% change in degree of dissociation $= \dfrac{0.4 - 0.2}{0.2} \times 100 = 100$

27. 4

Sol. $NH_2^-(\text{Bent}), OH_2(\text{Bent}), SO_2(\text{Bent}), SF_2(\text{Bent})$.

28. 4

Sol. 34 = (28 – 2) + n × 2

n = 4.

29. 15

Sol. $NaOH + HCl \longrightarrow NaCl + H_2O$

4	4	0	–
0	0	4	–

$[NaCl] = \dfrac{4}{30}$

$\Lambda_{eq} = \dfrac{k_1.1000}{N} = \dfrac{\left(2.2 \times 10^{-2} - 2 \times 10^{-2}\right) \times 1000}{4/30}$

= 15

30. 16

Sol. $\gamma = k[A]^{\alpha}[\beta]^{\beta}$

From Exp. I and III $\alpha = 2$

From Exp. III and IV $\beta = 0$

$k_1 = \dfrac{\gamma}{[A]^2} = \dfrac{4 \times 10^{-4}}{(0.1)^2} = 4 \times 10^{-2}\ L\,mol^{-1}\,sec^{-1}$

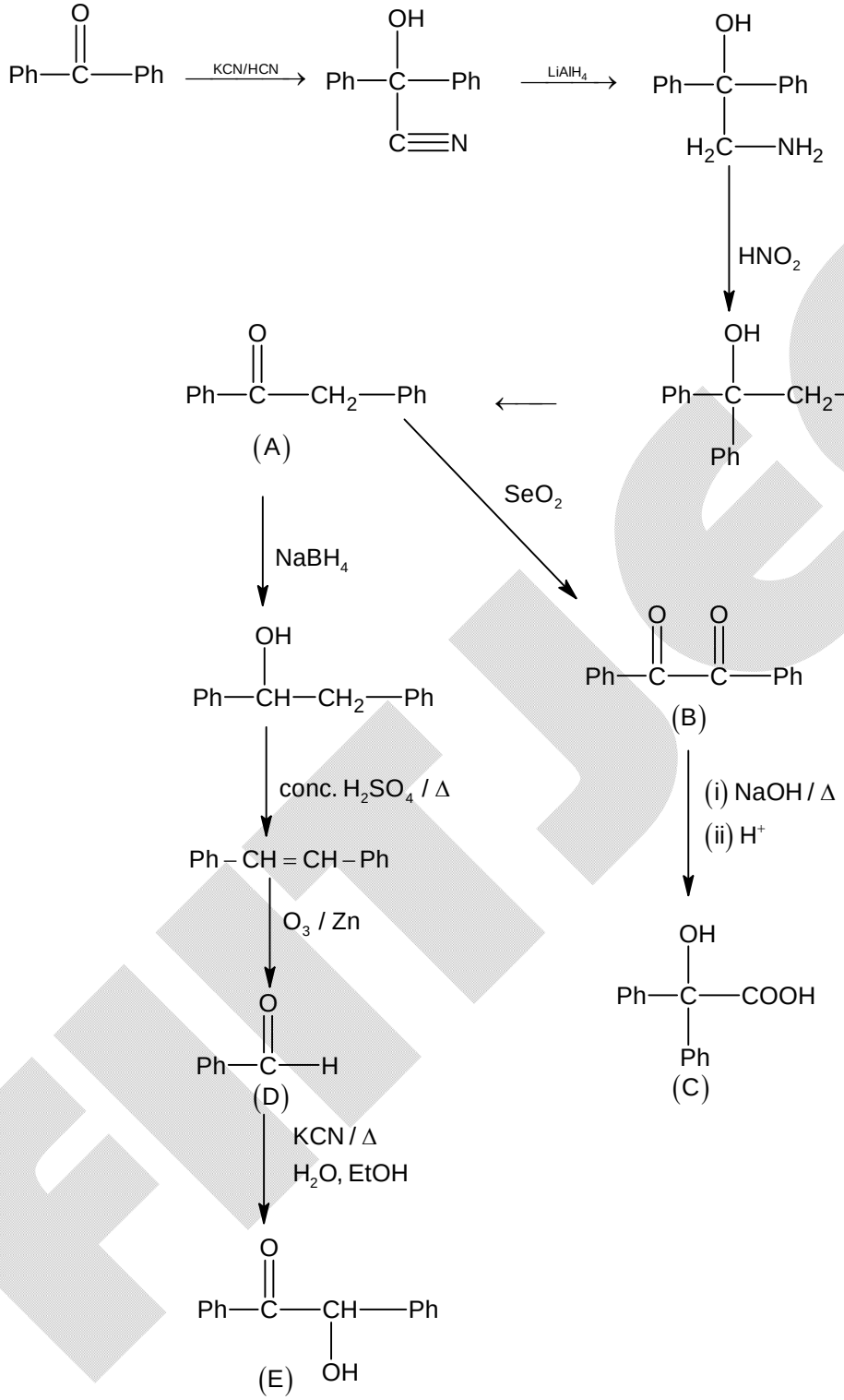
$\gamma = 4 \times 10^{-2} \times (0.2)^2 \times (0.35)^0 = 16 \times 10^{-4}\ Ms^{-1}$

SECTION – C

31. 9.00

32. 13.00

Sol. **(for the Q. No. 31 and 32)**



33. 4.00

Sol. $H_2CO_3 + NaOH \longrightarrow NaHCO_3 + Na_2CO_3$

10	15	0	0
0	0	5	5

$pH = x = 10 + \log\left(\frac{5}{5}\right) = 10$

After adding H_2SO_4

$NaHCO_3 + Na_2CO_3 + H^+ \longrightarrow H_2CO_3$

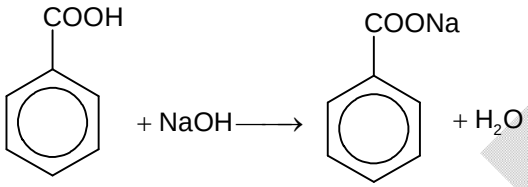
5	5	10	0
5	0	0	5

$pH = y = 6 + \log\left(\frac{5}{5}\right) = 6$

$x - y = 10 - 6 = 4$

34. 3.70

Sol.



0.2 mol	0.5×0.1 mol	0	
0.15	0	0.05	

$pH = pKa + \log\left(\frac{0.05}{0.15}\right)$

$= 4.2 - \log 3 = 3.7$

Mathematics

PART – III

SECTION – A

35. B

Sol. $\theta \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$, $\sin\theta \in \left(\frac{\sqrt{3}}{2}, 1\right)$ & $\sin^3\theta \in \left(\frac{3\sqrt{3}}{8}, 1\right)$

$\Rightarrow \quad \left(\sin\theta - \sin^3\theta\right) \in (0, 1)$ & $\sin\theta + \sin^3\theta > 1$

$\therefore \quad f(x) = \begin{cases} ax^3 - bx^2, & x \in Q \\ c - 11x, & x \notin Q \end{cases}$

$\therefore$ f(x) is continuous at x = 1, x = 2 & x = 3

$\therefore$ $ax^3 - bx^2 = c - 11x$ must have roots 1, 2 & 3

$\Rightarrow$ a = 1, b = 6, c = 6.

36. A

Sol. Let $\vec{r} = \hat{x} + y\hat{j} + z\hat{k}$

$x + y + 4z \le 14$; $x, y, z \in N$

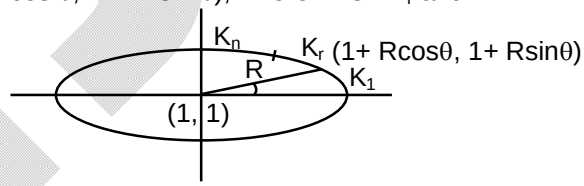
Number of triplets (x, y, z) = $1 + {}^6C_4 + {}^{10}C_8 = 61$.

37. C

Sol. $\frac{(x-1)^2}{a^2} + \frac{(y-1)^2}{b^2} = 1$

The coordinates of the point K_r will be $(1 + R\cos\theta, 1 + R\sin\theta)$, where R is AK_r and

$\theta = \frac{r\pi}{2n} \Rightarrow \frac{R^2\cos^2\theta}{a^2} + \frac{R^2\sin^2\theta}{b^2} = 1$



$\lim_{n\to\infty}\left(\frac{1}{n}\left(\sum_{r=1}^{n}\left(AK_r\right)^2\right)\right)$

$\Rightarrow \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{1}{\frac{\cos^2(r\pi/2n)}{a^2} + \frac{\sin^2(r\pi/2n)}{b^2}} = \int_0^1 \frac{a^2b^2}{b^2\cos^2\left(\frac{\pi}{2}x\right) + a^2\sin^2\left(\frac{\pi}{2}x\right)}dx$

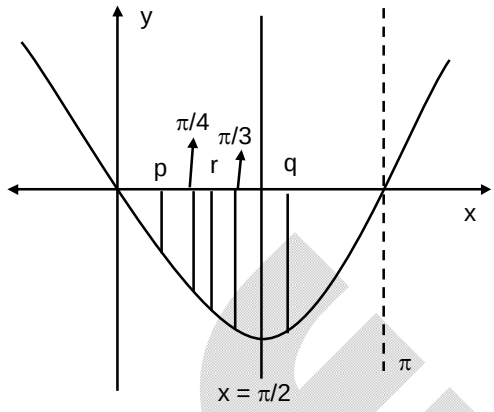
38. D

Sol. $(n-1)!n! - (n-2)!(n-1)!\ {}^{2n-2}C_1 = (n-1)!(n-2)!\{n(n-1) - (2n-2)\}$

$= (n-1)!(n-2)!\{n^2 - 3n + 2\}$

39. A, B

Sol. f(x) is symmetric about $x = \frac{\pi}{2}$ and g(x) is symmetric about $x = \pi$



also by observation: $\frac{\pi}{6} < p < \frac{\pi}{4} < r < \frac{\pi}{3} < q < \frac{2\pi}{3}$

$\Rightarrow \left|q - \frac{\pi}{2}\right| < \frac{\pi}{6} < \left|r - \frac{\pi}{2}\right| < \frac{\pi}{4} < \left|p - \frac{\pi}{2}\right| < \frac{\pi}{3}$

$\Rightarrow$ f(p) > f(r) > f(q) and g(p) > g(r) > g(p)

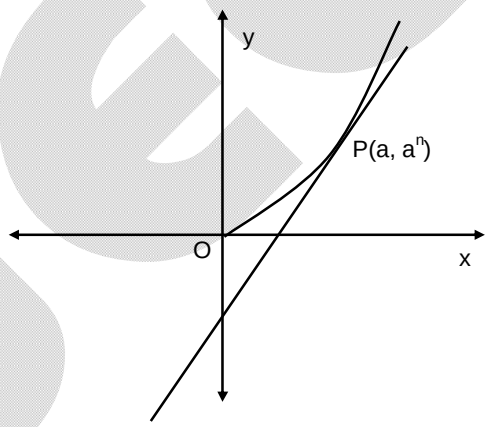
40. A, D

Sol. Slope of normal at $P(a, a^n)$ is $\frac{-1}{na^{n-1}}$

Equation of normal is $(y - a^n) = \frac{-1}{na^{n-1}}(x - a)$

Put x = 0 to get y-intercept as $b = a^n + \frac{1}{na^{n-2}}$

$$\Rightarrow \lim_{a\to 0} b = \begin{cases} 0, & \text{if } n < 2 \\ \frac{1}{2}, & \text{if } n = 2 \\ \infty, & \text{if } n > 2 \end{cases}$$



41. A, B

Sol. Slope of tangent of hyperbola xy = a at point (x, y) is

$\frac{dy}{dx} = -\frac{y}{x}$

Let the slope of tangent of curves which makes an angle of $\pi/4$ be $\frac{dy}{dx}$

$$\frac{\frac{dy}{dx} + \frac{y}{x}}{1 - \frac{y}{x}\frac{dy}{dx}} = \pm 1 \Rightarrow \frac{dy}{dx} + \frac{y}{x} = \pm 1 \mp \frac{y}{x}\frac{dy}{dx}$$

$$\frac{dy}{dx}\left(1 \pm \frac{y}{x}\right) = \pm 1 - \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{1 - \frac{y}{x}}{1 + \frac{y}{x}} \text{ or } \frac{-1 - \frac{y}{x}}{1 - \frac{y}{x}}$$

SECTION – B

42. 24

Sol. Let $z_1 = e^{i\theta_1}$, $z_2 = 2e^{i\theta_2}$, $z_3 = 3e^{i\theta_3}$

$\because z_1 + z_2 + z_3 = 0 \Rightarrow z_1^3 + z_2^3 + z_3^3 = 3z_1z_2z_3$

$\Rightarrow \cos 3\theta_1 + 8\cos 3\theta_2 + 64\cos 3\theta_3 = 24\cos(\theta_1 + \theta_2 + \theta_3)$

43. 7

Sol. Let $B = A - \frac{1}{3}A^2 + \frac{1}{9}A^3 + ... \infty$

$-\frac{AB}{3} = -\frac{A^2}{3} + \frac{1}{9}A^3 - \frac{1}{27}A^4 + ...$

$\Rightarrow \left(I + \frac{A}{3}\right)B = A \Rightarrow B = \frac{1}{3}(3I + A)^{-1}A$

Now, $B = \frac{1}{8}\begin{bmatrix} -\frac{1}{3} & 1 & 1 \\ 1 & -\frac{1}{3} & 1 \\ 1 & 1 & -\frac{1}{3} \end{bmatrix}$

44. 13

Sol. $f'(x) = \sec^2 x\left(3\tan^2 x + 2a\tan x + b\right)$

For $f'(x) \geq 0$ in $x \in R - (2n+1)\frac{\pi}{2}$

$D \leq 0 \Rightarrow a^2 \leq 3b, P(E) = \frac{4}{9}.$

45. 4

Sol. $A = \left(4 - \int_0^1 (f(y) - 1)\,dy\right) + \left|4 - \int_{-1}^0 (1 - f(y))\,dy\right| = \frac{9}{2}$

46. 4

Sol. Equation of the line L is $x + y - 1 + \lambda z = 0$, $x - y - 2 + \mu(y + z - 3) = 0$
As line passes through (1, 1, 1) so the value of λ will be –1 and $\mu = -2$

47. 2

Sol. $X + iy = \frac{a_r + 2i}{(a_r)^2 + 4}$

$X = \frac{a_r}{a_r^2 + 4}$, $Y = \frac{2}{a_r^2 + 4}$

$X^2 + Y^2 = \frac{1}{2}Y$

$X^2 + Y^2 - \frac{1}{2}Y = 0$

So all points lies on circle $x^2 + y^2 - \frac{y}{2} = 0$

$r = \frac{1}{4}$

Area of regular octagon = $8\left(\frac{1}{2} \times r^2 \times \sin\left(\frac{2\pi}{8}\right)\right) = 8 \times \frac{1}{2} \times \frac{1}{16} \times \frac{1}{\sqrt{2}} = \frac{1}{4\sqrt{2}}$

SECTION – C

48. 12.00

Sol. $(x^2 + x + 1)\, P(x - 1) = (x^2 - x + 1)\, P(x)$ ..... (1)

Since, $x^2 + x + 1$ and $x^2 - x + 1$ have no factor

Let $p(x) = (x^2 + x + 1)\, \phi(x)$

From (1), $\phi(x - 1) = \phi(x)$

$\Rightarrow \phi(x) = \phi(x + 1) \Rightarrow$ periodic

A polynomial is periodic if it is constant

$\Rightarrow P(x) = k(x^2 + x + 1)$, $P(1) = 3$

$\Rightarrow P(x) = x^2 + x + 1$

Now, $\int_0^1 \tan^{-1}\left(\frac{2x}{1+P(x^2)}\right)dx = \int_0^1 \tan^{-1}\left(\frac{2x}{1+x^4+x^2+1}\right)dx$

$= \int_0^1 \left\{\tan^{-1}(x^2+x+1) - \tan^{-1}(x^2-x+1)\right\}dx = \int_0^1 \left\{\tan^{-1}\left(\frac{1}{1+x^2-x}\right) - \tan^{-1}\left(\frac{1}{1+x^2+x}\right)\right\}dx$

$= \int_0^1 \left\{\tan^{-1}x - \tan^{-1}(x-1) - \tan^{-1}(1+x) + \tan^{-1}x\right\}dx$

$\int_0^1 \tan^{-1}\left(\frac{2x}{1+P(x^2)}\right)dx + \int_0^1 \tan^{-1}(1+x)\,dx = 3\left[\frac{\pi}{4} - \frac{1}{2}\ln 2\right] = \frac{3}{4}[\pi - \ln 4] \Rightarrow k = 12$

49. 2.00

Sol. $I_n = 2^n \int_0^1 (1+x^4)^n\, dx$

$\Rightarrow I_n = 2^n \left[x \cdot (1+x^4)^n\Big|_0^1 - \int_0^1 x \cdot n \cdot 4x^3 (1+x^4)^{n-1}\, dx\right]$

$\Rightarrow I_n = 2^n \cdot 2^n - 2^n \cdot 4n \int_0^1 (1+x^4-1)(1+x^4)^{n-1}\, dx$

$\Rightarrow I_n = 4^n - 4n \cdot 2^n \int_0^1 \left\{(1+x^4)^n - (1+x^4)^{n-1}\right\}dx$

$\Rightarrow I_n = 4^n - 4n \cdot I_n + 8n \cdot I_{n-1}$

$\Rightarrow (1+4n) I_n = 4^n + 8n I_{n-1}$

By comparison, $4^n = 8n$

$\Rightarrow n = 2$

50. 8.00

51. 17.00

Sol. **(for Q. 50 and 51).**

Angle bisector passes through focus in first question, angle bisector will be same in second question thus B will be (–1, 4)